

Floating dielectric slab optical interconnection between metal-dielectric interface surface plasmon polariton waveguides

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Abstract: A simple and effective optical interconnection which connects two distanced single metal-dielectric interface surface plasmon waveguides by a floating dielectric slab waveguide (slab bridge) is proposed. Transmission characteristics of the suggested structure are numerically studied using rigorous coupled wave analysis, and design rules based on the study are given. In the wave-guiding part, if the slab bridge can support more than the fundamental mode, then the transmission efficiency of the interconnection shows strong periodic dependency on the length of the bridge, due to the multi-mode interference (MMI) effect. Otherwise, only small fluctuation occurs due to the Fabry-Pérot effect. In addition, light beating happens when the slab bridge is relatively short. In the wave-coupling part, on the other hand, gap-assisted transmission occurs at each overlapping region as a consequence of mode hybridization. Periodic dependency on the length of the overlap region also appears due to the MMI effect. According to these results, we propose design principles for achieving both high transmission efficiency and stability with respect to the variation of the interconnection distance, and we show how to obtain the transmission efficiency of 68.3% for the 1mm-long interconnection.

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1. Introduction

As the amount of raw digital data and information increases enormously, the demand for data computing and processing speed grows rapidly and continuously. Recently, there has been much research focusing on ways to overcome obstacles in developing next-generation information processing devices, which are mainly related to the reduction of interconnection time delays [1]. Among the scarce solution trials, the optical data transfer scheme based on silicon photonics has been researched extensively, since it is able to eliminate most of the interconnection time delays in all conventional electronic circuits: device-to-device, chip-to-chip, layer-to-layer and board-to-board interconnection time delays [2-4]. This scheme, however, has a problem in miniaturization for successful integration in VLSI because of the diffraction limit in classical optics. Restricting the dimension of any optical data line to at least larger than $\lambda/2$, where λ is the wavelength of the light data carrier, the diffraction limit makes trouble with the integration of the optical data transfer scheme into modern VLSI circuits which have signal line widths of only few tens of nanometers.

To deal with this diffraction limit problem, a field named plasmonics has recently emerged as a possible solution. Plasmonics, a branch of nanophotonics, is emerging to deal with this

problem recently. Plasmonics is a research field that studies nano-scaled optical phenomena related to surface plasmons (SPs). SPs are quasi-particles that result from the quantization of plasma oscillation of electron gas in a metal surface. Coupling to SPs, electromagnetic wave forms localized field enhancement or propagating surface lightwaves called surface plasmon polaritons (SPPs) [5]. Since it is possible for SPP to achieve subwavelength confinement, plasmonics has attracted many researchers looking for a scalable and integratable optical data transfer/process schemes. Accordingly, numerous SPP-based plasmonic devices such as SPP-based waveguides, directional couplers and splitters, diffractive modulation devices have been introduced [5-9].

However, the SPP has a major drawback as an information carrier; it generally suffers from high propagation loss. The propagation length of the SPP in simple metal-air interface is, for example, 20 μm for silver, under the 500 nm of operation wavelength. Furthermore, this attenuation is fundamentally inevitable and increases as the modal size or wavelength of the SPPs decreases [6]. Despite all the efforts to increase the propagation length of SPPs [10], it seems SPPs need much way to overcome this problem. Therefore, it is seen as a reasonable time to adopt a different, new alternative; a hybrid scheme: using plasmonic devices for intra-chip data transfer and conventional optical devices for relatively long-distanced connections such as chip-to-chip, layer-to-layer and board-to-board connections.

Because of the reasons mentioned above, many researchers have studied the hybrid optical interconnection scheme recently [11-14]. Hochberg *et al.* implemented both plasmonic waveguide and dielectric ridge waveguide together using conventional semiconductor fabrication techniques [11]. Veronis and Fan proposed a multi-sectional taper based compact couplers between the dielectric slab waveguide and the metal-dielectric-metal (MDM) plasmonic waveguide and optimized it with a microgenetic optimization algorithm [12]. Sun and Zeng examined the coupling of the MDM waveguide mode and the single metal-dielectric interfaced SPP (MD-SPP) waveguide mode [13]. Ditlbacher *et al.* studied the coupling characteristics in the dielectric/metal multilayered film, both numerically and experimentally [14].

Aforementioned studies have paid attention to focusing on either implementation or evaluation of the proposed structures. Nevertheless, it is worth studying to suggest some principle of design rules based on the physical characteristics of the structure, despite the difficulties in a theoretical approach. We, therefore, aim to give not only an effective interconnection structure, but also a set of key design rules supported by detailed theoretical study.

In this paper, the interconnection structure for connecting two distanced air-based MD-SPP waveguides, which are simplified representations of plasmonic devices, is suggested and evaluated to give successful rules of design. In particular, we are going to show that it is possible to achieve both high transmission and robustness with respect to variation of the distance between two MD-SPP waveguides with the dielectric slab waveguide floating over them; over 70% of transmission rate and less than 5% of fluctuation of the transmission rate, under the appropriately designed structure. For this purpose, we first define the structural parameters in the suggested structure, and then we evaluate the transmission efficiency with respect to these parameters and analyze the underlying physical features. The transmission efficiency is calculated by the rigorous coupled-wave analysis (RCWA) method [15-18]. We finally conclude the paper, suggesting some important design principles of the structure based on the features studied in main section.

The rest of the paper is organized as follows. In Section 2, detailed descriptions of the interconnection structure and simulation methods are given. Physical explanations to the simulation results are covered in Sections 3 and 4. In particular, each covers the transmission characteristics mainly related to the wave-guiding (Section 3) and the wave-coupling (Section 4). Finally, concluding remarks are given in Section 5, with the suggestion of successful principles of design based on the results covered in Sections 3 and 4.

2. Descriptions of the interconnection structure and simulation method

Figure 1(a) illustrates a schematic diagram of the suggested structure. The dielectric slab waveguide (slab bridge) floats over two air-based MD-SPP waveguides with certain length of overlapping and thickness of air-gap. We assume that the operation wavelength is 632.8 nm and the dielectric constants of the metallic block, air, and the slab bridge are $\epsilon_m (-19 + j0.1)$, $\epsilon_a (= 1)$, and $\epsilon_d (= 2.25)$, respectively. We adopt the characteristics of the metallic blocks from the empirical optical constants of silver measured by Johnson and Christy [19]. In this structure, the incoming SPP mode of an MD-SPP waveguide, which propagates along the z -direction (input mode, 1), couples with the optical mode of the slab bridge (slab mode, 2) in the overlapping region, and the slab mode couples with the SPP mode of the other MD-SPP waveguide (transmitted mode, 3) on the other side of overlapping region. During the mode coupling processes, reflection and diffractive radiations occur at each edge of the overlapping regions (4, 5).

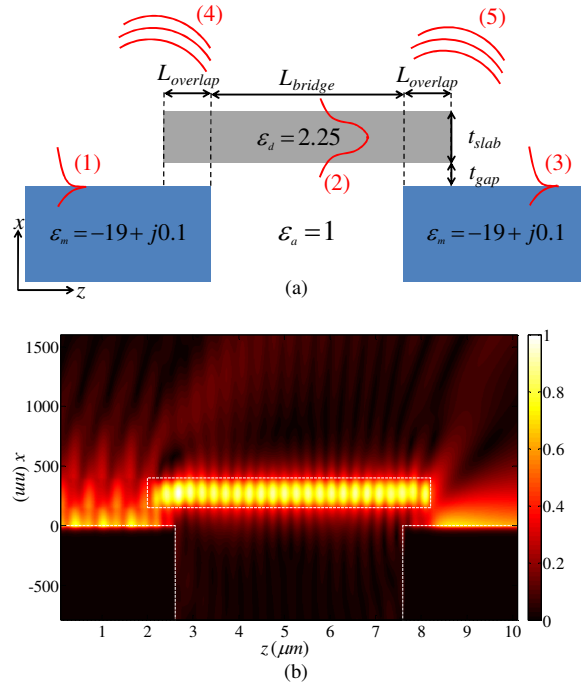


Fig. 1. (a) Schematic diagram of a floating dielectric slab interconnection between two air-based single metal-dielectric interface SPP waveguides. Each red drawing with number indicates corresponding eigenmode or scattered wave in given geometry. (b) [Media 1](#) The amplitude distribution of the magnetic field (H_y) of the floating dielectric slab interconnection; with $t_{gap}=150 \text{ nm}$, $t_{slab}=250 \text{ nm}$, $L_{overlap}=600 \text{ nm}$, and $L_{bridge}=5 \text{ μm}$, respectively.

There are four structural parameters which determine the amount of energy of each wave component; the thickness of the air-gap, the thickness of the slab bridge, the length of each overlapping region, and the length of the slab bridge; each denoted by t_{gap} , t_{slab} , $L_{overlap}$, and L_{bridge} , respectively. In this structure, only TM polarized electromagnetic waves – which are consisted of E_x , E_z , and H_y – can be applied because the MD-SPP waveguide can support only TM polarized modes. In Fig. 1(b), the magnetic field (H_y) amplitude distribution of the given structure is presented; with $t_{gap}=150 \text{ nm}$, $t_{slab}=250 \text{ nm}$, $L_{overlap}=600 \text{ nm}$, and $L_{bridge}=5 \text{ μm}$. In advance, the movie linked to Fig. 1(b) shows the propagation of electromagnetic waves in the same structure as Fig. 1(b), depicting how the proposed structure operates as an optical interconnection between two MD-SPP waveguides.

To study the transmission characteristics theoretically, we calculate the transmission efficiency (T_{eff}) defined as the intensity ratio of the transmitted mode to the input mode for various conditions of the structural parameters. For the calculation, the structural parameters are categorized into two parts: the parameters related to the wave-guiding such as L_{bridge} and t_{slab} , and those related to wave-coupling such as $L_{overlap}$ and t_{gap} . T_{eff} is evaluated with respect to the two parameters in each category and the following results are deeply investigated.

For numerical simulation, the RCWA method [15-18] is employed. RCWA is the most well-known and well-developed Fourier-based modal analysis technique for Maxwell equations [18]. It constructs any possible mode and scattered wave component inside the structure in terms of Fourier series, and determines the amplitude of each by matching the boundary conditions. Since Fourier series is the most successful approximation of any function in the mean squared error point of view, RCWA can construct the mode profile in the structure not only easily, but also accurately even if the structure is complex and multi-layered. This makes RCWA the most suitable technique for analyzing modes inside the structure. Furthermore, RCWA is also useful in studying mode-coupling engaged problems like our given problem because the coupling coefficients between modes are automatically obtained during the boundary condition matching process [17, 18]. In fact, T_{eff} is obtained by calculating the absolute squared value of the coupling coefficient of the input mode to the transmitted mode. In this paper, we conduct each numerical calculation under the following configuration: the total number of x -direction Fourier spatial harmonics is 200, and the size of computation cell period (in the x -direction) is 10 μm , which shows a reasonable convergence in every situation we evaluate.

3. Transmission characteristics related to the wave-guiding process

In Fig. 2(a), the transmission efficiency T_{eff} is shown as a function of L_{bridge} and t_{slab} ; with L_{bridge} varying from 1000 μm to 1010 μm , t_{slab} from 0 nm to 1000 nm , when $L_{overlap}$ and t_{gap} are fixed as 600 nm and 100 nm , respectively. We divide the result into two regimes with respect to t_{slab} : the single mode regime and the multi mode regime. These regimes are determined by the cutoff thickness of the 1st excited mode. The cutoff thickness of the M th excited mode ($t_{c,M}$) is the maximum value of t_{slab} which makes the slab bridge cut the M th excited mode off. It is given by [20]

$$t_{c,M} = \frac{M \lambda_0}{2NA}, \quad (1)$$

where λ_0 is the operation wavelength in free space ($=632.8$ nm), and NA is the numerical aperture of the slab bridge, given by $NA = \sqrt{\epsilon_d - \epsilon_a}$. Therefore, the cutoff thickness of the 1st excited mode is given as $t_{c,1}=282$ nm , according to Eq. (1).

In the single mode regime ($0 < t_{slab} < t_{c,1}$), where the slab bridge can only hold the fundamental mode, there are two remarkable characteristics in the variance of T_{eff} . First of all, T_{eff} increases monotonically until t_{slab} increases to around 200 nm , and afterwards slowly decreases, as shown in Fig. 2(b). Furthermore, this tendency fits with the squared overlapping integral value of the SPP mode and the fundamental slab mode. This indicates that the transmission efficiency is proportional to the squared overlap integral between the input mode and the slab mode, which is result of the coupled-mode theory since the coupling coefficient between two independent waveguides depends mostly on the overlap integral of two waveguide modes [21, 22].

Another remarkable characteristic is that there is a periodic fluctuation that regards L_{bridge} with a 250 nm period, shown in Fig. 2(c) (red solid line). This periodic fluctuation is the consequence of the Fabry-Pérot resonance in the slab bridge, which clearly appears in Fig. 1(b) and Fig. 2(d). Recently, Valle *et al.* suggested a picture that describes the resonance in nano-strip structures [23]. According to the picture, forwarding and reflecting modes in the

slab bridge interfere with each other, making an interference pattern which consequently affects the transmission efficiency. This resonance condition is given by

$$L_{\text{bridge}} \frac{2\pi}{\lambda_0} n_{\text{eff}} = m\pi, \quad (2)$$

where n_{eff} is the effective index of the slab mode, and m ($=1,2,3,\dots$) is the phase-matching condition for constructive interference. The period of the resonance, therefore, can be calculated by using Eq. (2): $L_1 = \lambda_0/2n_{\text{eff}} = 247 \text{ nm}$. This value agrees well with Fig. 2(c), which confirms the validity of the picture.

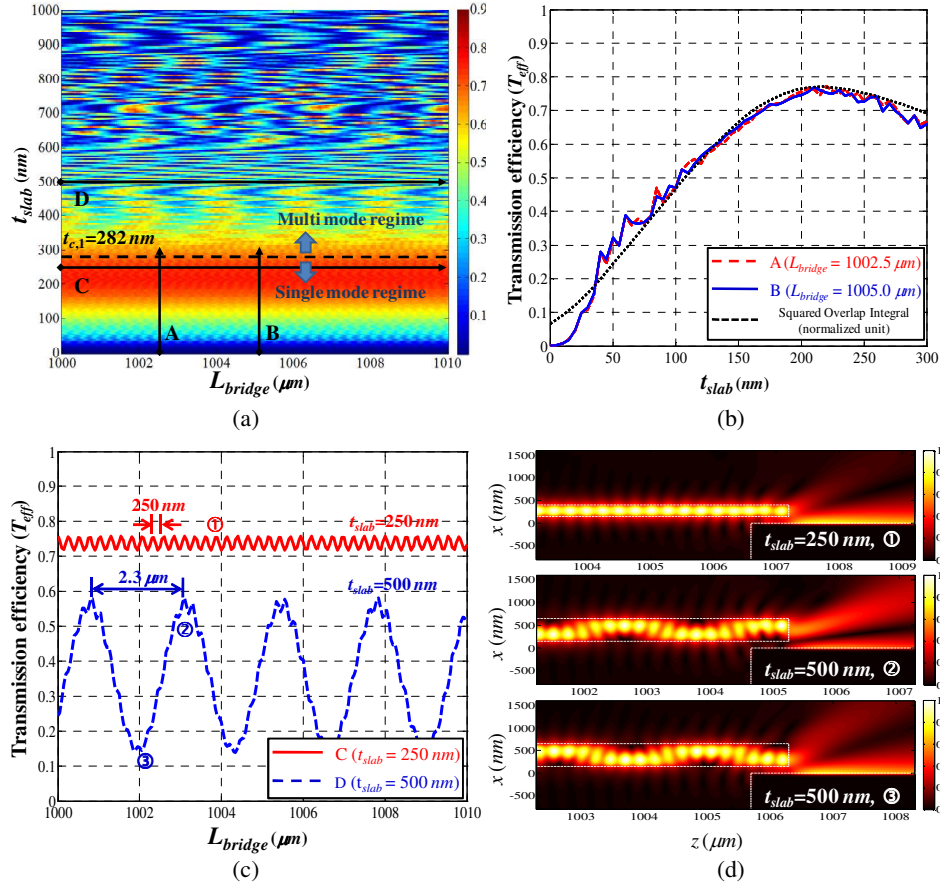


Fig. 2. (a) The transmission efficiency (T_{eff}) as a function of the length of the slab bridge (L_{bridge}), and the thickness of the slab bridge (t_{slab}). Each black arrowed line marked with capitals in the figure indicates the sweeping interval of the plot in (b) and (c), and the dashed line indicates the boundary of the single mode and multi mode regimes. (b) T_{eff} as a function of t_{slab} , for a given L_{bridge} . The red dashed line and the blue solid line correspond to the two different cases of slab bridge length denoted by A and B in (a), and the black broken line corresponds to the normalized-to-fit value of the squared overlap integral of the SPP mode and the fundamental slab mode. (c) T_{eff} as a function of L_{bridge} , for a given t_{slab} . The red solid line and the blue dashed line correspond to the single mode case (C) and the multi mode (D) case, respectively. (d) Amplitude distributions of the magnetic field at the out-coupling part of the interconnection. Each corresponds to the case of $t_{\text{slab}} = 250 \text{ nm}$ (①), the minimum (②) and maximum transmission (③) with $t_{\text{slab}} = 500 \text{ nm}$. Throughout the figures, all other structural parameters are fixed as $L_{\text{overlap}} = 600 \text{ nm}$, and $t_{\text{gap}} = 150 \text{ nm}$.

The transmission characteristics of the multi mode regime ($t_{\text{slab}} > t_{c,1}$) are, however, totally different from those of the single mode regime. First, T_{eff} varies with respect to L_{bridge} , with a

period and amplitude much larger than that of the Fabry-Pérot fluctuation effect, shown in Fig. 2(b) (blue dashed line). These periodic and rapid variations are consequences of the interference of all possible modes in the slab bridge, called multi-mode interference (MMI). Figure 2(d) illustrates the magnetic field distributions when this effect occurs. Because the amount of energy coupled to a high order mode increases as we increase t_{slab} , this interference gives more affect to the field distribution at the end of the slab bridge, causing high variation to T_{eff} , as shown in Fig. 2(a). When two slab modes interfere with each other, as in Fig. 2(b), the period of this effect can be determined by the condition of the constructive interference of two different waves, which is given by

$$L_{bridge} \frac{2\pi}{\lambda_0} (n_{eff,0} - n_{eff,1}) = 2m\pi, \quad (3)$$

where $n_{eff,0}$ and $n_{eff,1}$ are the effective indices of the fundamental and 1st excited modes of the slab bridge, respectively, and m ($=1,2,3,\dots$) is the condition for the zero phase difference between two modes. The calculated value is $L_2 = \lambda_0 / (n_{eff,0} - n_{eff,1}) = 2.334 \mu m$, which agrees well with the simulation result. Second, T_{eff} varies in a complex manner as t_{slab} increases. This occurs mainly because of two reasons; one is the decrease of n_{eff} of the slab mode, and the other is the interference with additional high order slab modes related with Eq. (1). Especially, the former one consequently decreases the optical path length (OPL) of the slab bridge, defined by

$$OPL = n_{eff} \cdot L, \quad (4)$$

where n_{eff} is the effective index, and L is the length of the travelling wave [24]. From Eq. (4), it is shown that the variation in t_{slab} causes similar modulation to that occurred when L_{bridge} varies.

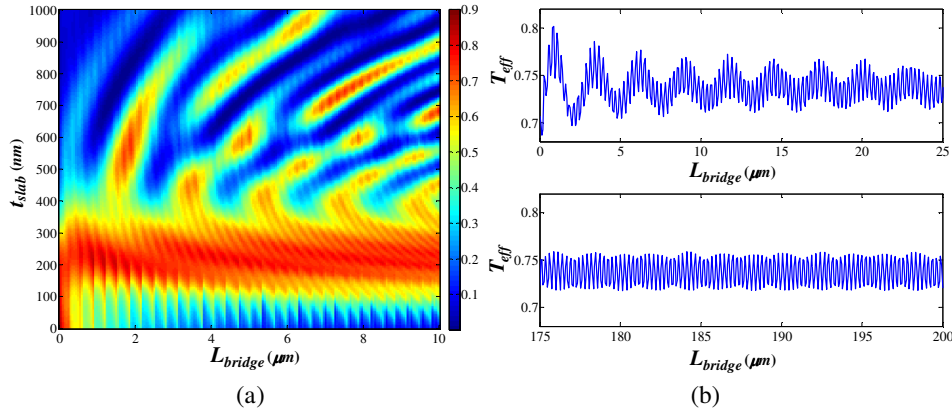


Fig. 3. (a) T_{eff} as a function of L_{bridge} and t_{slab} , when the sweeping interval of L_{bridge} and t_{slab} is 0~10 μm and 0~1000 nm, respectively. (b) T_{eff} as a function of L_{bridge} , in the case of a relatively short slab bridge (upper figure), and the long slab bridge (lower figure). In both cases, t_{slab} is 250 nm. Throughout the figures, all other structural parameters are set as $L_{overlap}=600$ nm and $t_{gap}=100$ nm.

On the other hand, this picture does not seem to agree with the simulation result when the slab bridge is short. As shown in Fig. 3(a), an unobvious interference pattern appears even though the slab bridge has only the fundamental mode. A similar phenomenon has been reported recently, in the study of the tunneling of SPPs across an interruption in the metallic film waveguide [25]. The same analogue can be used to explain this case. The interference pattern of the short slab bridge can be ascribed to the fact that the air-traveling scattered wave

components do not fully vanish and beat with the slab modes at the end of the slab bridge where they couples to the output mode. This pattern decays as L_{bridge} increases, as shown in Fig. 3(b), which is obvious since the air-traveling wave components continuously lose their energy as they travel.

4. Transmission characteristics related to the wave-coupling process

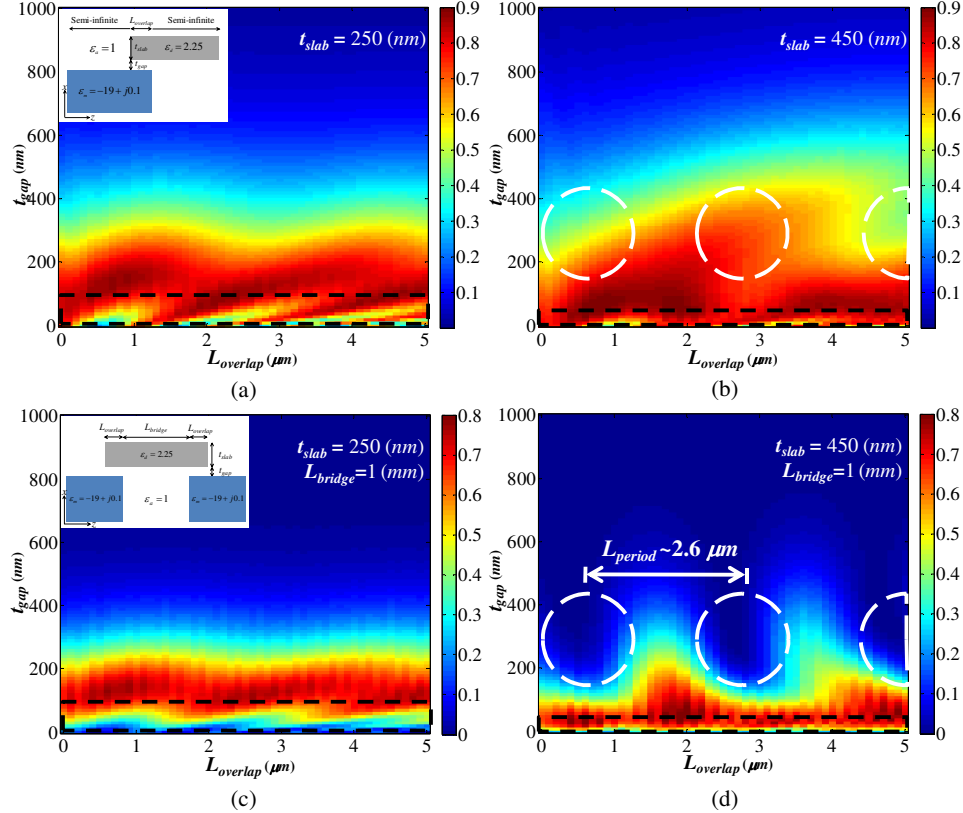


Fig. 4. T_{eff} as a function of $L_{overlap}$ and t_{gap} for the single wave-coupling structure only and the whole interconnection structure. We show T_{eff} for the single wave-coupling structure when (a) $t_{slab}=250$ nm and (b) $t_{slab}=450$ nm. The inset of (a) represents the schematics of the single wave-coupling structure. Like above, we illustrate T_{eff} for the whole structure when (c) $t_{slab}=250$ nm and (d) $t_{slab}=450$ nm. The inset of (c) represents the schematics of the whole interconnection structure. Through (c) and (d), the length of the slab bridge is held to 1 mm. The circles in (b) and (d) are marked in same positions. Throughout all figures, the low-gap interference is marked with dotted rectangles.

At first, we evaluate T_{eff} of the single wave-coupling structure, which is illustrated in the inset of Fig. 4(a), as a function of $L_{overlap}$ and t_{gap} , with the following intervals; 0 μm to 5 μm for $L_{overlap}$, and 0 nm to 1000 nm for t_{gap} . The result is illustrated in Figs. 4(a) and 4(b). In the result, strong interference occurs for relatively small t_{gap} in both cases, marked with a rectangle. To explain this low gap interference, understanding of the mode hybridization process is needed. According to Refs. 10, 26 and 27, hybridization of the SPP mode and the optical mode, called “plasmon hybridization”, induces two mutually coupled modes each called a symmetric hybrid mode and an antisymmetric hybrid mode. This process is called mode splitting. Figure 5(a) gives the mode profiles of those hybrid modes. For the effective indices of these two hybrid modes, it is known that the symmetric hybrid mode has a lower effective index than that of the SPP mode and the antisymmetric hybrid mode has a higher effective index than that of the optical mode [10, 26]. Furthermore, these differences increase

as two waveguides approach each other, because the interaction strength of polarization charges in the slab bridge waveguide and SPs in the MD-SPP waveguide increases [26, 27]. Figure 5(b) clearly shows the aforementioned features. As a consequence, these hybrid modes interfere with each other to make such periodic fluctuation. For the case of Fig. 4(a), antisymmetric and symmetric hybrid mode has the effective index of $n_{\text{eff,asym}}=1.5758$ and $n_{\text{eff,sym}}=1.0005$, respectively, and corresponding calculated period is $1.099 \mu\text{m}$. It agrees with the simulation result in the above. Interestingly, in Fig. 4(a), this periodic fluctuation seems to be weakened with the increasing of t_{gap} , and the coupling does not seem to depend strongly on L_{overlap} when t_{gap} is around 100 nm . This is because the decrease of the effective index of the antisymmetric hybrid mode makes it easier to be coupled with the SPP mode, by decreasing the difference between the effective index of the antisymmetric hybrid mode and the SPP mode. It makes the coupling to the antisymmetric hybrid mode increase and the symmetric hybrid mode decrease. As a result, the amount of interference between two hybrid modes decrease, which reduces the periodic fluctuation with respect to L_{overlap} .

On the other hand, when we evaluate the transmission characteristics of the whole structure, additional complexity associated with L_{bridge} comes into the action. Figures 4(c) and 4(d) illustrate T_{eff} of the whole interconnection structure with respect to L_{overlap} and t_{gap} , with the same intervals of Figs. 4(a) and 4(b). There is no remarkable distinction between Figs. 4(a) and 4(c). However, a remarkable distinction appears between Figs. 4(b) and 4(d). In detail, T_{eff} at the circled areas of Fig. 4(b) seem to extinct in Fig. 4(d). Furthermore, these extinctions occur periodically, with the period being approximately $2.6 \mu\text{m}$. This periodic pattern implies that strong interference between some wave components exists. We have already discussed a similar phenomenon and its underlying feature: the MMI effect. The interference pattern in Fig. 4(d) is caused by interference between the two modes inside the overlapping region. In this case, each two of the modes has effective index of $n_{\text{eff},1}=1.3924$ and $n_{\text{eff},2}=1.1562$, respectively. The period of the MMI pattern, therefore, can be calculated by Eq. (3) and its value is $2.68 \mu\text{m}$. This calculated period matches well with the simulation result in Fig. 4(d). It clearly shows that the MMI effect dictates when the slab bridge can support multiple modes, and consequently degrades the stability with respect to L_{bridge} .

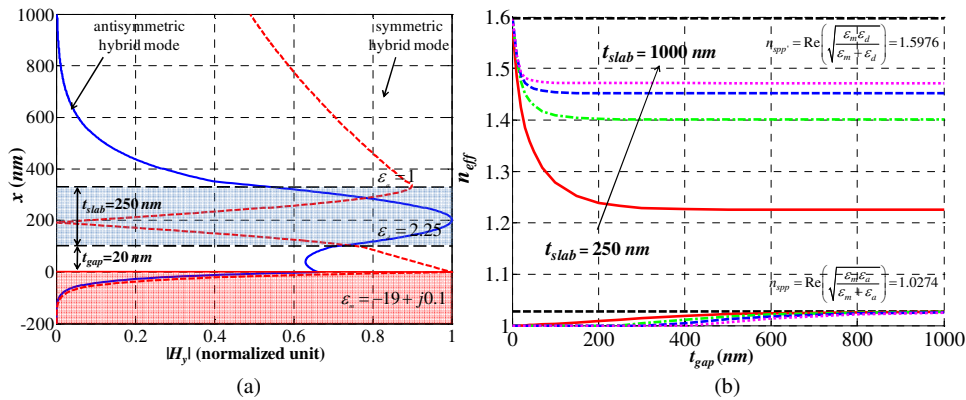


Fig. 5. (a) The amplitude distributions of the magnetic field (H_y) of the symmetric (red dashed line) and the antisymmetric hybrid (blue solid line) modes. Each red and blue shaded area corresponds to the metal surface and dielectric slab, respectively, and the white region corresponds to the air. (b) The effective indices of the hybrid modes with respect to t_{gap} , for $t_{\text{slab}}=250 \text{ nm}$ (red solid line), 500 nm (green dash-dot line), 750 nm (blue dashed line), and 1000 nm (magenta dotted line). The effective indices of the SPP mode of the metal-air and metal-dielectric single interface are given in black dashed lines. Throughout the figures, $L_{\text{overlap}}=600 \text{ nm}$ and $L_{\text{bridge}}=1 \text{ mm}$.

With the exception of the interference effects in the overlapping region, some interesting phenomena are revealed in the transmission characteristics related to t_{gap} and t_{slab} . Figures 6(a) and 6(b) present T_{eff} as a function of t_{slab} and t_{gap} , with the intervals of both 0 nm to 1000 nm. The results show the different characteristics according to the condition whether t_{slab} is in the single mode or the multi-mode regime, just as the results of section 3. In the single mode regime, maximum transmission is achieved when the air gap is within 100–200 nm, as shown in Figs. 6(a) and 6(b). Furthermore, T_{eff} rather diminishes to zero when two waveguides are close together. It does not agree with our intuition in mode coupling – the coupling efficiency should be proportional to the overlap integral of the two modes – since the coupling efficiency of this case is not maximum at $t_{gap}=0$, even though the overlap integral is maximum. This gap-assisted transmission is plausible, however, if we consider the following two features: the effective index elevation of the antisymmetric hybrid mode, and the field distribution of the symmetric hybrid mode.

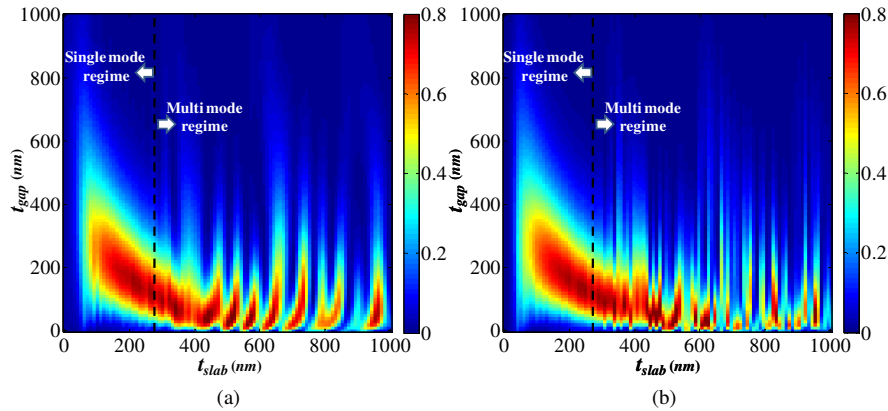


Fig. 6. T_{eff} as a function of t_{slab} and t_{gap} , for both short and long slab bridges. The length of the slab bridge is (a) $L_{bridge}=10\ \mu m$ (short) and (b) $L_{bridge}=1\ mm$ (long). For both cases, the length of the overlapping region is $L_{overlap}=600\ nm$ each. The dashed line in each figure divides the single mode regime and the multi-mode regime.

Let us firstly consider the effective index elevation effect of the antisymmetric hybrid mode. As mentioned in the previous paragraph, the effective index of the antisymmetric hybrid mode grows, caused by the polarization hybridization process. Furthermore, the decrease of t_{gap} aggravates this effective index elevation, as illustrated in Fig. 5(b). Since the transmittance and reflectance depend on the difference of the refractive index, the large difference of the effective indices between the SPP mode and the antisymmetric hybrid mode causes a remarkable reflection at the interface of the overlapping region. Moreover, due to the polarization charge distribution at the slab bridge and the metal surface, the magnetic field distribution of the symmetric hybrid mode has both positive and negative values at once, whereas that of the antisymmetric one has only positive values, as shown in Fig. 5(a). Furthermore, the value of the magnetic field of the symmetric hybrid mode crosses zero at the middle of the slab bridge. Therefore, the modal overlapping of the symmetric hybrid mode and the fundamental slab mode is remarkably smaller than that of the antisymmetric hybrid mode and the fundamental slab mode. Hence, coupling to the symmetric hybrid mode is relatively weak even though the difference of the effective indices with the SPP mode is significantly small. Consequently, all these features work together and bring the gap-assisted transmission effect, as shown in Figs. 6(a) and 6(b).

Interestingly, the gap-assisted transmission diminishes as t_{slab} grows. It originates from the weakening of mode splitting. Since the mode characteristic of the hybrid mode is much like the slab mode when the slab is relatively thick [7], the difference of effective indices

between the slab mode and the hybrid mode gets smaller as the slab bridge thickens, as shown in Fig. 5(b). Hence the difference in the effective index by mode splitting does not give a significant difference in the reflectance any more. In addition, the enlargement of the slab thickness gives a larger modal size to the slab mode, which gives a significant increment in the overlap integral. Furthermore, this overlap integral grows dramatically as the waveguides approach each other, illustrated in Fig. 7. It causes a strong mode coupling for the slab bridge with large thickness.

In the multi-mode regime, on the contrary, totally different characteristics can be seen in the results. First, high T_{eff} regions, which look like a series of spikes, evidently appear, as illustrated in Fig. 6(a). Moreover, the increasing of L_{bridge} seemingly complicates this pattern so that the result of the long slab bridge case in Fig. 6(b) rather looks like randomly distributed peaks. The difference between the two figures in Fig. 6 is, actually, somewhat similar with the difference between Fig. 3(a) and Fig. 2(a). In fact, these are tied together with the common underlying feature; OPL. The increasing of t_{slab} makes the OPL increase, and it brings the same effect of increasing both $L_{overlap}$ and L_{bridge} . Since the slab bridge can contain more than two modes, this OPL increase brings intensity modulation at the end of the slab bridge, consequently making T_{eff} vary like so. Of course, there also exists the effect of increasing the number of modes which can be interfered with. These effects are aggravated as L_{bridge} increases, because the increment of OPL, which concerns about the degree of interference, is simply proportional to L_{bridge} , according to Eq. (4).

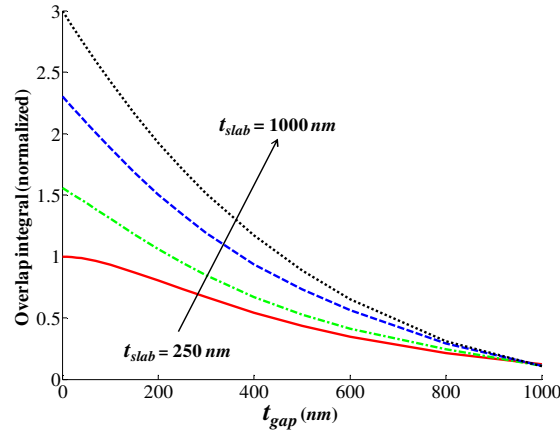


Fig. 7. Overlap integrals of the SPP mode and the slab mode with respect to t_{gap} , for $t_{slab}=250$ nm (red solid line), 500 nm (green dash-dot line), 750 nm (blue dashed line), and 1000 nm (black dotted line). Every overlap integral value is normalized with respect to the maximum overlap integral in the case of $t_{slab}=250$ nm.

5. Conclusion

This paper has suggested an effective structure for interconnecting two distanced MD-SPP waveguides by floating a dielectric slab waveguide on the edge of each MD-SPP waveguide. Numerically calculating T_{eff} in terms of various structural parameters with RCWA, the transmission characteristics inside the structure are also shown. In particular, we studied this interconnection structure from two points of view; the aspect of variation in the wave-guiding process, and that in the wave-coupling process. In the wave-guiding part, we showed that when t_{slab} is in the multi-mode regime, T_{eff} varies largely by period with respect to L_{bridge} , compared to the effect of the Fabry-Pérot which always occurs in the slab bridge with any t_{slab} . It degrades the stability of interconnection, since it makes T_{eff} too sensitive to the variation in L_{bridge} . Furthermore, when the slab bridge is too short to radiate scattered wave components out, light beating comes into action and makes T_{eff} fluctuate with respect to L_{bridge} . In the

wave-coupling part, on the other hand, we showed that mode hybridization gives the gap-assisted transmission effect. There also exists severe stability degradation caused by the MMI effect when the slab thickness is in the multi-mode regime or the air gap is almost zero.

From these observations, we propose some principles when designing the parameters of the suggested floating slab interconnection structure. First of all, thickness of the slab bridge should never be larger than $t_{c,1}$, in order to achieve good stability with respect to the length of the slab bridge. Secondly, to obtain high transmission, it is recommended to make the proper thickness of the gap between the MD-SPP waveguide and the slab bridge waveguide, which is 100~150 nm under the given geometry and the 633 nm of operating wavelength. Finally, the length of the overlap region does not have to be strictly regulated unless the thickness of the slab bridge is in the multi-mode regime.

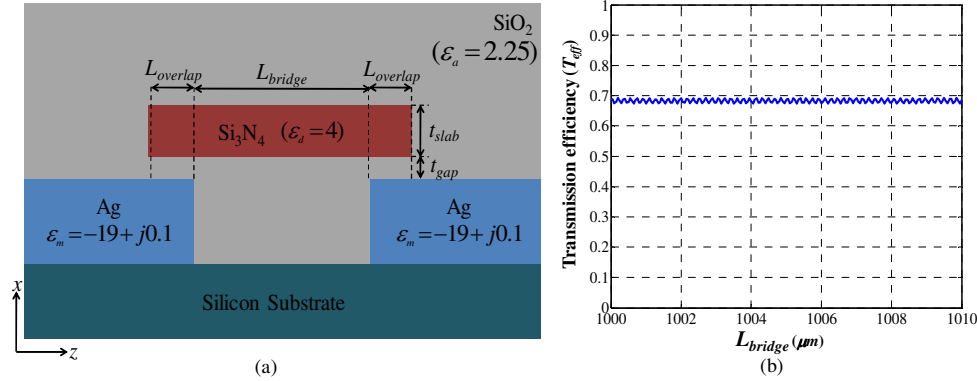


Fig. 8. (a) Schematic diagram of the practical example of a floating dielectric slab interconnection between two air-based single metal-dielectric interface SPP waveguides. (b) T_{eff} of (a) as a function of L_{bridge} , when values of other structural parameters are $t_{slab}=230$ nm, $L_{overlap}=600$ nm, and $t_{gap}=75$ nm.

By applying these principles, we design the proposed structure by letting $t_{slab}=250$ nm, $t_{gap}=150$ nm, and $L_{overlap}=600$ nm and obtained average transmission efficiency as 70 % for a 1 cm-long slab bridge ($L_{bridge}=1$ cm). The obtained transmission efficiency showed only 2 % of peak-to-peak fluctuation. The floating gap might be controlled by the size of beads that might be inserted between the dielectric slab and metal. We also apply the design principles to suggest the practical example which can be implemented by silicon fabrication technology. In Fig. 8(a), we illustrate the schematic diagram of the implementable interconnection structure. There are two differences between this case and the case illustrated in Fig. 1(a). First of all, the material which fills the entire structure in Fig. 8 is fused silica (SiO_2 , $\epsilon_r=2.25$), whereas Fig. 1(a) is simply free space ($\epsilon_r=1$). Another difference is the material of the slab bridge waveguide: silicon nitride (Si_3N_4 , $\epsilon_r=4$) is used in Fig. 8, whereas fused silica is used in Fig. 1(a). We adopt the material constants of the dielectric materials used in this structure from Ref. 28. Letting $t_{slab}=230$ nm, $t_{gap}=75$ nm, and $L_{overlap}=600$ nm, we obtain average transmission efficiency as 68.3% for a 1mm-long slab bridge ($L_{bridge}=1$ mm) and 1.6 % of the peak-to-peak fluctuation with respect to the variation in L_{bridge} , which clearly shows that the interconnection structure is also effective in practical use. The detailed simulation result is illustrated in Fig. 8(b). In this simulation the waveguide mode and SPP mode are calculated considering the dielectric constants of fused silica and silicon nitride.

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